

DecodeLabs Internship

Project 3 Report

Project Title: Customer Segmentation using Unsupervised Learning

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Objective: Segment customers into meaningful groups using unsupervised machine learning techniques to support business decision-making.

1. Introduction

Customer segmentation helps businesses understand different purchasing patterns and customer behavior. In this project, an e-commerce dataset was analyzed and grouped using K-Means clustering. Principal Component Analysis (PCA) was applied to reduce dimensionality while preserving most of the information. The quality of clusters was evaluated using the Elbow Method and Silhouette Score.

2. Dataset Overview

The dataset contains transactional information such as Product, Quantity, Unit Price, Total Price, Payment Method, Order Status, Referral Source, Items in Cart, and Date. Additional features including Year, Month and Day of Week were extracted from the Date column for analysis.

3. Methodology

Step	Description
Data Cleaning	Removed duplicates and handled missing values using statistical imputation.
Feature Engineering	Extracted Year, Month and DayOfWeek from Date.
Preprocessing	Applied StandardScaler and One-Hot Encoding.
Dimensionality Reduction	Applied PCA to retain approximately 95% of the variance.
Clustering	Used K-Means algorithm for customer segmentation.
Model Selection	Used Elbow Method and Silhouette Score to determine the optimal number of clusters.
Business Analysis	Generated customer personas based on cluster characteristics.

4. Results & Interpretation

The clustering model successfully grouped customers with similar purchasing behaviour. PCA enabled efficient visualization of customer groups in two dimensions. The Elbow Method and Silhouette Score were used together to identify an appropriate number of clusters. Each cluster can be interpreted as a unique customer persona, enabling targeted marketing strategies and personalized recommendations.

5. Tools & Libraries

Python, Pandas, NumPy, Scikit-learn, Matplotlib, PCA, K-Means, Jupyter Notebook.

6. Conclusion

This project demonstrates a complete unsupervised learning workflow for customer segmentation. After preprocessing and feature engineering, PCA reduced the dimensionality of the data and K-Means identified customer groups. The resulting segments can help businesses improve customer understanding, marketing campaigns, product recommendations and strategic decision-making.